

Zone.

AN OFFLINE, INFRASTRUCTURE-FREE DISASTER SIGNAL NETWORK

Proposal cum Report — submitted for incubation support to the Atal Incubation Centre, Sikkim Manipal Institute of Technology (AIC-SMUTBI)

Prepared by	Live product	Status	Date
Team Zone · Dept. of Computer Science & Engineering, SMIT	zone-thezone.vercel.app	Working native Android app · validated on real hardware	October 2026

Executive Summary

Zone is a native Android application that lets ordinary smartphones form a working communication network with zero cellular signal, zero Wi-Fi, and zero internet — the exact condition that follows a major earthquake, flood, or landslide across the Himalayan Northeast. Phones talk directly to each other over Bluetooth Low Energy, each carrying a trapped person's 31-byte status signal a little further, hand to hand, until it reaches a responder.

The product's core idea is one no comparable tool implements: every signal a phone sends declares exactly when it intends to send the next one. When a phone stops transmitting, the network does not just see silence — it can tell whether that silence was *expected* (a dying battery, announced in advance) or *unexpected* (a promise broken, meaning the device — and likely the person holding it — was lost). When a cluster of devices in one area goes silent together, the system infers that the area itself was destroyed, at a known minute, before any responder arrives. We call this the Dead Man's Packet.

This is not a concept deck. Zone is a working, signed Android release, validated through real two- and three-phone Bluetooth tests in airplane mode, backed by 135 automated unit tests, with a companion online situational-awareness dashboard already live. We are seeking AIC-SMUTBI's support to take it from a validated prototype to a field-piloted product for Sikkim's disaster-response agencies, with a clear path to the wider Northeast.

1 The Problem

In the first 72 hours after a major disaster, the same infrastructure that would normally carry a call for help is usually the first thing to fail. Cell towers lose power or backhaul; the networks that remain are saturated by everyone else trying to do the same thing. Sikkim and the wider Himalayan Northeast are structurally exposed to this: seismically active terrain, flood- and GLOF-prone river systems, and a road network that itself fails under the same event that triggers the disaster, isolating communities exactly when coordination matters most.

Two gaps follow directly from this, and they are the two problem statements this project is built against (Avinya 2026, PS2 and PS5):

Ground-zero blackout	A trapped or stranded person with a working phone and no signal has no way to be heard. Satellite phones are scarce, need a sky view, and are not something a trapped civilian is carrying.
Post-disaster information fog	Incident command at the edge of a blackout zone has no reliable way to tell a totally collapsed block from a merely flooded one until a team physically walks in — wasting the exact hours survival probability depends on.

Existing mesh-messaging apps (Bridgefy, Briar, goTenna) are built for two conscious people chatting — they have no concept of an expected next message, so a contact going quiet means nothing to them. Zone is not a messaging app. It is structured telemetry, and that distinction is what makes silence itself a usable signal.

2 The Solution

A single Android APK, three role-specific screens chosen once on first launch, no accounts, no setup beyond granting Bluetooth and Location:

Citizen	For the person in danger. Opens straight into broadcasting — no menus, no login. Works even if the phone is buried or the person is unconscious, because status is sensor-derived by default; buttons only refine it.
Responder	A triage-ranked list of everyone heard, sorted by who is in the worst situation and who is about to go silent — not by arrival time. A proximity bar ('Dig Here') guides the final few metres by signal strength, built for one gloved hand, not a map.
Map / EOC	A severity grid for a command post or projector, where a cell that has gone completely silent is drawn with its own unmistakable visual language and an exact timestamp — deliberately never allowed to look like a cell that simply has no coverage.

Every packet is exactly 31 bytes — the legacy Bluetooth advertising payload limit, chosen deliberately so the protocol never needs a fallback path. It carries a device identifier, a position delta, a status and severity, battery level, barometric altitude (basement vs. rooftop, to $\pm 1\text{--}2$ m, where GPS is dead), and the field the whole system is built on: **next_expected_tx** — how many seconds until this device intends to speak again.

“A life in 31 bytes, and the system knows the difference between a phone that went quiet and a phone that was destroyed.”

Delivery does not depend on any one link holding. Every phone that hears a packet stores it and rebroadcasts it to whoever it meets next — an epidemic, content-addressed store-carry-forward relay. In live testing, a signal from a phone with its own Bluetooth switched off still reached a third phone two hops away, carried physically in a second phone's pocket.

3 What Is Already Built and Validated

This is deliberately the longest section of this proposal: AIC-SMUTBI is being asked to back engineering that already works, not an idea.

Capability	Status	Evidence
Dual-PHY BLE mesh (1M + Coded), store-carry-forward relay	Built, validated on hardware	2-hop delivery confirmed with source Bluetooth off, on three physical phones and, independently, on two phones in a live session, both in airplane mode.
Dead Man's Packet — expected vs. unexpected silence	Built, validated on hardware	A phone powered off at healthy battery was correctly flagged UNEXPECTED_SILENCE with a wall-clock timestamp; the same phone at critical battery was correctly flagged EXPECTED, with no false escalation.
CELL_LOSS — simultaneous mass-silence detection	Built, validated against a 500-node simulation	Reproduced against a replayed casualty curve from a real regional disaster (22→95→469→626 over 72 h); renders as a distinct, timestamped hole on the map, never flat no-data.
Battery-adaptive duty cycle (PHY + interval ladder)	Built, validated on hardware	Confirmed switching between Coded (long-range) and 1M PHY live, tracking battery level in real time on a physical phone.
Barometric relative altitude + drowning escalation	Built, validated on hardware (3-phone test)	Floor-change detection within roughly ± 1 m of ground truth; rising-water status correctly promoted a device to the top of the triage list.

Capability	Status	Evidence
Severity, headcount and status-phrase self-reporting	Built, validated live	Set on one phone, correctly decoded and displayed on a second phone over real Bluetooth, same test session.
Mesh-wide emergency alert (siren, torch, Bluetooth-name beacon)	Built, validated live	An alert issued on one phone produced a full-screen takeover, siren and a renamed Bluetooth beacon on a second phone, received over the live mesh.
Per-responder ECDSA P-256 signing (anti-forgery upgrade)	Built, unit-tested; not yet load-bearing	19 dedicated unit tests (sign/verify, tamper, wrong-key, unprovisioned); layered on top of today's shared-key authentication without breaking it, pending a field trial before activation.
Automated test suite	135 JVM unit tests, zero Android dependency in the core logic	Runs in continuous integration on every push; covers the packet codec, the relay store, the silence evaluator, and the cryptography independently of any device.
Online companion dashboard (Phase 1, pre-disaster / connected use)	Built and live	Real-time public hazard feeds (USGS earthquakes, NASA FIRMS fires, NASA EONET, GDACS disaster alerts) and public traffic cameras on a 3D/satellite map, plus the ability to load a phone's own exported field data as a precise, reverse-geocoded pin.

Full field-test protocols, including an unmeasured range survey and multi-phone soak testing, are documented and ready to run as device access scales up — exactly the kind of hardware-lab support this proposal is asking AIC-SMUTBI for.

4 Technical Architecture

Platform

Native Kotlin + Jetpack Compose, minSdk 26 / targetSdk 34 — no cross-platform framework, because peripheral BLE advertising is where hybrid frameworks fail.

Transport

An abstracted ReportTransport interface with three implementations — real BLE, an in-process simulator (demo and test insurance), and a file-based sneakernet fallback — so every piece of logic above the radio runs identically regardless of which one is active.

Zero network, by rule

No servers, no analytics, no downloaded map tiles anywhere in the offline app. It is built to work in airplane mode or it does not ship.

Identity & trust

A per-install device key today; a shared responder key authenticates official RESOLVE/ALERT messages; the built per-responder ECDSA signing layer is the scoped next step past that, with no new runtime dependency — only platform-native java.security APIs.

Deployment

A single signed release APK, distributed from one Vercel-hosted link that also carries the offline control-room viewer and the Phase-1 online dashboard — one link to share, verified byte-identical from source to device on every release.

5 Market and Deployment Path

The near-term market is institutional, not consumer: State and District Disaster Management Authorities, NDMA-aligned emergency operations centres, NGOs running relief operations, and search-and-rescue teams who today have no tool that turns a dead phone into a data point. Sikkim is the natural first deployment — compact, high disaster-risk, and the home of this project — with a direct path to Meghalaya and the wider Northeast once a pilot is proven, given the shared terrain and infrastructure profile across the region.

Zone's own adoption curve does not depend on everyone having the app before a disaster strikes: every phone that *does* have it immediately extends the mesh for every phone around it, and a single responder team carrying provisioned phones into a blackout zone is already a usable deployment on day one.

6 Roadmap

Horizon	Milestone
0–3 months	Field pilot with one Sikkim disaster-response unit; multi-phone (5+ device) soak and range testing across real terrain; Play Store listing and compliance documentation.
3–6 months	Broker camera/sensor data access with Gangtok Smart City (GSCDL) for the Phase-1 dashboard; activate the ECDSA signing layer as the primary trust mechanism, after field validation; formal engagement with Sikkim SDMA.
6–12 months	Expand the pilot to a second Northeastern state (Meghalaya); publish a public after-action report from real deployments; pursue NDMA-level visibility.

7 What We Are Asking AIC-SMUTBI For

Device lab access

A pool of varied Android handsets (different chipsets, different Bluetooth stacks) to replace today's two-to-three-phone testing with the real-world device diversity a public deployment will meet.

Institutional introductions

A path to Sikkim SDMA and Gangtok Smart City (GSCDL) for an official pilot and for the camera-data access the Phase-1 dashboard is designed to use.

Mentorship

Guidance on compliance, data-handling policy, and the security review the responder-signing upgrade needs before it can be trusted as the primary authentication layer.

Incubation & funding support

To be scoped jointly with AIC-SMUTBI, against the pilot plan above — primarily device procurement, field-trial logistics, and Play Store / compliance costs.

In return, Zone offers AIC-SMUTBI a project that is unusually far along for its stage: a working, tested, signed product, built and validated in Sikkim, for a problem Sikkim itself faces directly.

8 Team

Team Zone — Department of Computer Science & Engineering, Sikkim Manipal Institute of Technology. *[Team member names and roles to be added here.]*

Live product: [zone-thezone.vercel.app](#) · Install card (QR): [zone-thezone.vercel.app/install.html](#) · Source: [github.com/Arunodoy18/TheZone](#)